

A multi-feature selection fused with investor sentiment for stock price prediction

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ARTICLE INFO

Keywords:

Feature selection
Investor sentiment
Price prediction
Deep learning

ABSTRACT

The stock market data has the characteristics of high dimensionality and multiple noise, and the investment behavior of investors is easily influenced by emotions, which poses a great challenge to stock price prediction. To improve the accuracy of stock price prediction, this study proposes a combined modeling approach based on multiple feature selection algorithms and incorporating investor sentiment. First, we collected stock trading data of the Chinese A-share market from 2018 to 2022, and three types of investor sentiment data sourced from social media, Internet news and newspaper news. Then, we used five feature selection algorithms to select dozens of important features from hundreds of features in the stock trading data. Based on three types of investor sentiment data, five sentiment indicators were constructed and included in the subsequent feature selection along with the previously selected important features. Finally, five deep learning models were used to predict stock prices using feature sets with sentiment indicators. A total of 1030 stocks from 10 industries such as pharmaceutical and biological, leisure service, food and beverage were selected for the experiment. The results show that in 10 industries, LSTM-CNN-Attention model has the best effect on stock price prediction, and after incorporating the sentiment indicator constructed by the principal component, the effect of the model is significantly improved, and the performance is the best in 7 industries. This method explores a new way of stock price prediction by integrating investor sentiment, and can provide further reference for the research of stock price intelligent prediction.

1. Introduction

The stock market is often regarded as a barometer of a nation's or region's economic and financial activities. Fluctuations in stock prices have consistently been a focal point for investors, with the ability to forecast market trends being crucial for informed decision-making. (Badea et al., 2019) However, predicting stock market movements is a complex and challenging task, primarily due to the high-dimensional nature of the data, the presence of significant noise, and redundancy, compounded by the influence of various factors such as global economic conditions, political events, corporate performance, and public sentiment. (Yang and Wang, 2022) Within the stock market, price volatility itself is a nonlinear, dynamic, and unstable process, inherently containing noise, both large and small, which significantly impacts stock prices. (Nti et al., 2019) Furthermore, stock price forecasting not only relies on technical analysis and data evaluation but is also profoundly

affected by market sentiment. Investors' emotional responses often influence price fluctuations, thereby exerting substantial effects on predictions of stock prices. (Srijiranon et al., 2022) In recent years, the rapid development of artificial intelligence (AI) technologies has introduced new possibilities for stock price prediction. Leveraging robust models and extensive historical data, AI can assist investors in better understanding market trends, identifying potential opportunities, and making more informed investment decisions. (Cui et al., 2023).

In the early stages of stock prediction, statistical methods such as Autoregressive Integrated Moving Average (ARIMA) (Babu and Reddy, 2015) and Grey Prediction Models (Tian et al., 2020) were commonly employed. However, when dealing with large volumes of stock data characterized by high volatility, traditional statistical methods often failed to produce satisfactory forecasting results. Nowadays machine learning and deep learning methods, including Support Vector Machines (SVM), Artificial Neural Networks (ANN), Long Short-Term Memory

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<https://doi.org/10.1016/j.eswa.2025.127381>

Received 4 February 2025; Received in revised form 19 March 2025; Accepted 20 March 2025

Available online 26 March 2025

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Networks (LSTM), and their hybrid models, have been increasingly applied in stock market forecasting (Hafezi et al., 2015; Kim, 2003; Rather, Agarwal, & Sastry, 2015; Xiong & Lu, 2017; Lee & Um, 2019). Wang and Wang (2016) obtained stock comment information from social media and pre-processed it into emotion vector. Based on SVM-based model, they used efficient market hypothesis to predict stock prices. Sisodia et al. (2022) proposed a prediction model based on LSTM for stock market prediction using the nifty fifty 50 dataset and achieved an accuracy of 83.88%.

The use of deep neural networks for stock market forecasting generally falls into two broad categories. The first method utilizes only historical stock data as input features for prediction, while the second method incorporates additional relevant features, including technical indicators, social sentiment, news sentiment, and other external factors, to enhance the prediction process.

Regarding the first approach, stock market data often possesses high dimensionality, containing a large number of features. Relying solely on one or a few features can, to some extent, capture the time series characteristics of stock prices, but its limitations are also quite apparent. High-dimensional data not only increases the computational complexity but can also lead to model overfitting, thereby affecting the accuracy and stability of the prediction results. Feature selection methods address this issue by evaluating and selecting features that have a significant impact on prediction, while removing redundant and irrelevant features. This process reduces the data's dimensionality, alleviates computational burden, and enhances the model's generalization capability. Haq et al. (2021) employed three feature selection techniques—logistic regression, Support Vector Machines (SVM), and Random Forest—to compute the importance of features, ultimately selecting 20 features. The results indicated that a combination of features selected by multiple, non-overlapping techniques provided higher accuracy for the predictive model compared to those selected by individual feature selection methods. Similarly, Kou et al. (2021) applied four feature selection methods to determine the optimal subset of features for bankruptcy prediction in small and medium-sized enterprises, highlighting the importance of the feature selection process in improving the performance of predictive models.

Regarding the second approach, with the gradual exploration of sentiment data and other external information sources, an increasing number of studies have begun to integrate social media data, news sentiment, and analysis reports—along with other forms of sentiment and socio-political event information—into stock market prediction models. Sentiment indicators reflect the emotional fluctuations of market participants, which often influence collective market behavior and, in turn, impact stock price movements. By combining sentiment data with stock trading data, models can better capture the irrational volatility of the market, thereby improving the accuracy of predictions. In a study by Lee et al. (2024), an ESG sentiment index extracted from news articles was integrated with technical indicators to predict the Standard & Poor's 500 index. The results demonstrated that the accuracy of the predictions was higher when ESG sentiment was taken into account. Similarly, Shankar et al. (2024) designed and developed an ensemble model that combined Long Short-Term Memory (LSTM) networks, ARIMA, and public sentiment extracted from Twitter, further validating that incorporating external features into predictive models can significantly enhance the accuracy of stock price forecasting.

To reduce the complexity of high-dimensional stock market data, this study employs diversified feature selection techniques for dimensionality reduction, while incorporating multi-source market sentiment factors and utilizing deep learning models to predict stock closing prices. The motivation behind this approach is as follows:

1) High Dimensionality and Noise in Stock Market Data: Stock market trading data typically contains a vast amount of feature information, such as historical stock prices, trading volumes, and technical indicators. As the volume of data increases, the feature space becomes highly dimensional. However, not all features are meaningful for stock

price prediction. Many features may be irrelevant to stock price movements or even contain noise, which not only increases the computational complexity of the model but can also lead to overfitting.

2) Impact of Multi-Source Investor Sentiment on Market Volatility: Investor sentiment is one of the key drivers of market volatility, particularly in today's age of rapidly flowing information and the widespread use of social media. The influence of sentiment fluctuations on stock prices has become more pronounced. Investor sentiment affects stock price movements not only directly through market news and financial reports, but also manifests through more complex irrational decision-making patterns via collective behavior. In recent years, social media, online news, and newspapers have become major sources of sentiment information. However, existing prediction methods predominantly focus on using historical trading data or single-source sentiment for forecasting, often neglecting the combined impact of multi-source sentiment factors.

3) Limitations of Existing Prediction Methods: Traditional stock price prediction methods typically rely on a single data source, such as historical trading data, technical indicators, or fundamental analysis. As a result, these methods are often inadequate when dealing with the complexities of dynamic market environments. A single data source fails to capture the multidimensional and dynamic information of the market, particularly when sentiment factors are not fully incorporated into the model. Moreover, many existing methods lack effective cross-industry validation across different markets and sectors, resulting in limited generalizability and scalability. Therefore, integrating multiple data sources—such as trading data and sentiment data—and applying advanced modeling techniques to effectively learn from these multidimensional features has become a crucial research direction for enhancing the performance of stock price prediction methods.

Based on the aforementioned motivations, the innovations of this approach are summarized as follows:

1) Integration of Multiple Feature Selection Algorithms: This study innovatively combines five different feature selection algorithms to address the high dimensionality and noise issues in stock market trading data, proposing a diversified feature selection framework. Traditional methods often rely on a single feature selection technique, which may lead to over-reliance on certain features or the omission of important information. For instance, some algorithms may excel at selecting highly correlated features, while others may perform better in removing noisy features. By integrating multiple algorithms, this research can uncover potential patterns in the data from different perspectives, thus avoiding biases that may arise from using a single method. This not only improves the accuracy of feature selection but also enhances the robustness of the model in dealing with complex data, thereby providing a more solid foundation for subsequent stock price prediction.

2) Fusion Modeling of Multi-Source Sentiment Indicators and Trading Data: This study introduces a stock price prediction method driven by multi-source sentiment indicators by combining investor sentiment data with traditional stock trading data. We extracted sentiment data from social media, online news, and newspaper articles, constructing five sentiment indicators that reflect market sentiment, and fused these sentiment signals with trading feature data. In this way, sentiment data not only supplements the irrational factors of market behavior as external information but also provides multidimensional support for stock price prediction. Traditional stock price prediction models often overlook the role of multi-source sentiment, whereas this research systematically examines the impact of multi-source sentiment on stock price fluctuations, filling the gap in existing research and improving the accuracy and applicability of the prediction model through feature fusion.

3) Multidimensional Validation of Deep Learning Models: We not only employed various deep learning architectures but also conducted detailed experimental validations on different feature combinations. In traditional methods, models typically rely on a single feature set or simple predictive structures. However, through comparative

experiments involving feature sets with and without sentiment indicators, we explored the practical role of sentiment factors in stock price prediction. Additionally, extensive cross-industry validation was carried out, selecting data from various sectors such as pharmaceuticals, biotechnology, and food & beverage industries, demonstrating the universal applicability and stability of the proposed method. Experimental results show that the incorporation of sentiment indicators significantly enhances prediction accuracy, particularly in cross-industry validation, thus opening new avenues for sentiment-driven stock price prediction and providing fresh perspectives and methods for future related research.

This study is divided into several key parts, systematically introducing the research methods and results. In the second section, previous work and its limitations are discussed. The third section introduces the feature selection method, sentiment indicator construction method, deep learning method and evaluation approach of this research. In section four, the performance of the proposed model is described, and the potential capability of multi-source sentiment indicators constructed based on multiple feature selection methods in stock prediction is proved through experiments, and the superiority of the LSTM-CNN-Attention model is illustrated. Finally, in section five, we summarize the results, discuss the limitations and look forward to future work.

2. Related works

2.1. Predicting stock prices with deep learning

Due to its high volatility, the problem of stock price prediction has consistently posed a significant challenge to the research community. However, recent advancements in deep learning models have opened new avenues for addressing such data from diverse perspectives. (Usmani and Shamsi, 2023).

Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM) are two representative architectures in the field of deep learning. Compared to other deep learning approaches, CNN excels in feature extraction and information mining. The research by Hoseinzadeh and Haratizadeh (2019) demonstrated the remarkable performance of CNN in stock time series forecasting. However, a standalone CNN model lacks memory capabilities and struggles to capture the underlying patterns in sequential data. In contrast, LSTM shows a significant advantage in handling the changing patterns of temporal data. Sezer et al.'s (2020) study analyzed various machine learning and deep learning models, including CNN, DBN, and LSTM, and concluded that LSTM, due to its clear model structure and superior performance with time series data, is the most commonly used model in stock market prediction. A comparative study by Pokhrel et al. (2022) evaluated CNN, LSTM, and GRU for predicting closing prices on the Nepal Stock Exchange, using Root Mean Square Error (RMSE) as a performance metric. The input sequences included stock prices, macroeconomic data, technical indicators, and sentiment scores from financial news. The results indicated that LSTM outperformed both CNN and GRU. Furthermore, Singh et al. (2023) combined the strengths of CNN and LSTM, utilizing fundamental market data and technical indicators to capture subtle fluctuations in stock market data. Their findings demonstrated the effectiveness of the CNN-LSTM hybrid model in stock selection and portfolio optimization.

2.2. Stock historical feature affects stock prices

Stock prediction, like other machine learning tasks, involves selecting appropriate features from a given dataset and utilizing them to train predictive models. Feature selection not only helps reduce irrelevant variables, computational costs, and overfitting issues but also enhances the predictive performance of the model. (Cai et al., 2012) Currently, three main feature selection methods are commonly used in stock market prediction: Filter methods, Wrapper methods, and Embedded

methods. Filter methods rank features based on their correlation with the underlying machine learning algorithm, selecting the top-ranked features as a preprocessing step for subsequent learning algorithms. (Urbanowicz et al., 2018) Pearson's correlation coefficient is a typical technique within Filter methods. Li et al. (2022) used Pearson's correlation coefficient to select features with a correlation of 0.5 as input data for a Broad Learning System (BLS) model, aimed at predicting the closing price for the next day. Based on 11 years of experimental data from the Shanghai Stock Exchange, their research indicated that combining Pearson's correlation coefficient with BLS outperformed ten other traditional machine learning methods in terms of performance. In Wrapper methods, feature selection is embedded within the machine learning algorithm's training process, with the goal of identifying the feature subset that provides optimal predictive performance. Recursive Feature Elimination (RFE) is a well-known Wrapper feature selection technique. Yuan et al. (2020) applied the RFE algorithm to extract importance scores from 60 features across ten different categories, selecting the top 80% of features, ultimately using 48 features to predict stock returns in the Chinese A-share market. Embedded methods integrate the advantages of both Filter and Wrapper methods by performing feature selection simultaneously within the model training process of an ensemble algorithm, thus making feature selection part of the training process itself. Random Forest, as an ensemble learning method, has been increasingly used for feature selection in recent years. In the study by Botunac et al. (2020), Random Forest was employed to assess the importance of 14 features for predicting the closing prices of Apple, Microsoft, and Facebook.

2.3. Investor sentiment affects stock prices

Investor sentiment is a critical factor in predicting stock trends, often treated as an external variable and incorporated as input into predictive models. (Hu et al., 2021) Srijiranan et al. (2022) combined Principal Component Analysis (PCA), Empirical Mode Decomposition (EMD), and Long Short-Term Memory (LSTM) networks, utilizing historical stock market data and news sentiment analysis to enhance stock market predictions. Their findings indicate that the hybrid model demonstrated superior predictive performance, with news sentiment analysis significantly improving the LSTM model's accuracy. Koukaras et al. (2022) collected tweet data from Twitter and StockTwits, along with financial data from Yahoo Finance, and applied seven machine learning algorithms to assess the impact of sentiment analysis on stock prediction accuracy. The results revealed that the model combining the VADER sentiment lexicon with support vector machines yielded the best performance. Deng et al. (2024) incorporated sentiment data from institutional, individual, and foreign investors as input features to develop prediction models for the price trends of the Shanghai Composite Index and the Shenzhen Component Index, alongside trading simulations. Their results showed that the proposed XGBoost-Bagging-NSGA-II model outperformed benchmark methods in terms of hit rate, cumulative returns, and maximum drawdown, demonstrating its effectiveness and sustainability as an intelligent tool.

Although existing studies have extensively explored stock market prediction methods, including traditional statistical models and machine learning/deep learning approaches, and have considered the direct impact of investor sentiment on stock price fluctuations, these studies often rely on overly simplistic prediction metrics that fail to account for multiple indicators such as market trading activity and financial performance. Furthermore, the investor sentiment indicators selected are often too narrow in scope. Therefore, this study proposes a hybrid model that combines multiple feature selection algorithms with investor sentiment to improve the accuracy of stock price predictions. The approach adopted in this research not only incorporates a wide range of technical indicators but also integrates investor sentiment data from social media, online news, and newspaper articles to capture subtle shifts in market sentiment.

3. Methodology

3.1. Overall process

The general flow of this study is shown in Fig. 1. First, stock historical data and sentiment indicator data are collected. The collected stock historical data is preprocessed, and then feature selection is performed. Five feature selection algorithms are selected to classify the features in the stock historical data into three categories of “Strong”, “Medium” and “Weak” according to their importance. The features in the historical stock data are classified into three categories according to their importance: “Strong”, “Medium” and “Weak”, and the important features are selected. The collected sentiment indicator data is divided into three categories, namely social media, Internet news and newspaper news, and based on these three sentiment indicator data, three independent sentiment indicators and two combined sentiment indicators are constructed. Finally, the deep learning model is applied to stock price prediction and Mean Absolute Error (MAE), Mean Square Error (MSE) and Root Mean Squared Error (RMSE) are used as evaluation criteria.

3.2. Feature selection

The goal of feature selection is to find the optimal subset of features, eliminate irrelevant or redundant features, and achieve the purpose of reducing the number of features, improving the accuracy of the model, and reducing the running time. According to the form of feature selection, it can be divided into three categories: filter, wrapper and embedded. Common filter feature selection algorithms include Pearson correlation coefficient, Maximum Information Coefficient (MIC), Analysis of Variance (ANOVA), relief and chi-square test, etc. (Huang et al., 2022) The most commonly used wrapper feature selection algorithms are Recursive Feature Elimination (RFE) and Null Importance, etc. The

most commonly used embedded algorithms are Lasso Regression and Classification Decision Tree, etc. By comparing different categories of feature selection algorithms, the following commonly used feature selection algorithms are screened:

(1) Pearson's correlation coefficient. The correlation between features is assessed by calculating the strength of linear correlation between variables, r , as shown in Equation 1:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{Y})^2}}$$

Where n is the number of samples, x_i and y_i are the i observation corresponding to the variables X and Y , and $\bar{X}$ and $\bar{Y}$ denote the sample mean of the two, respectively. r takes the range of $[-1, 1]$, when $r > 0$, the variables are linearly and positively correlated; when $r < 0$, the variables are linearly and negatively correlated; and when $r = 0$, there is no linear relationship between the variables.

Pearson can quickly identify linear correlation features such as opening price and trading volume, and achieve effective feature screening on some data, but it cannot capture the hysteresis effect of sentiment indicators, which needs to be combined with time series analysis.

(2) MIC. This algorithm not only measures the linear correlation between variables, but also detects the nonlinear correlation between variables. MIC is to form a scatterplot of two variables x and y in two-dimensional space to form a grid of $x \times y$, and then compute the value of mutual information $I(X, Y)$ under different segmentation scales, and the maximum value of mutual information is $\max\{I(x, y)\}$, which is normalized to the normalized value, and then obtain the normalized feature matrix through Eq 2. The normalized feature matrix is obtained by Eq 2, where $p(x, y)$ is the marginal probability density function of X

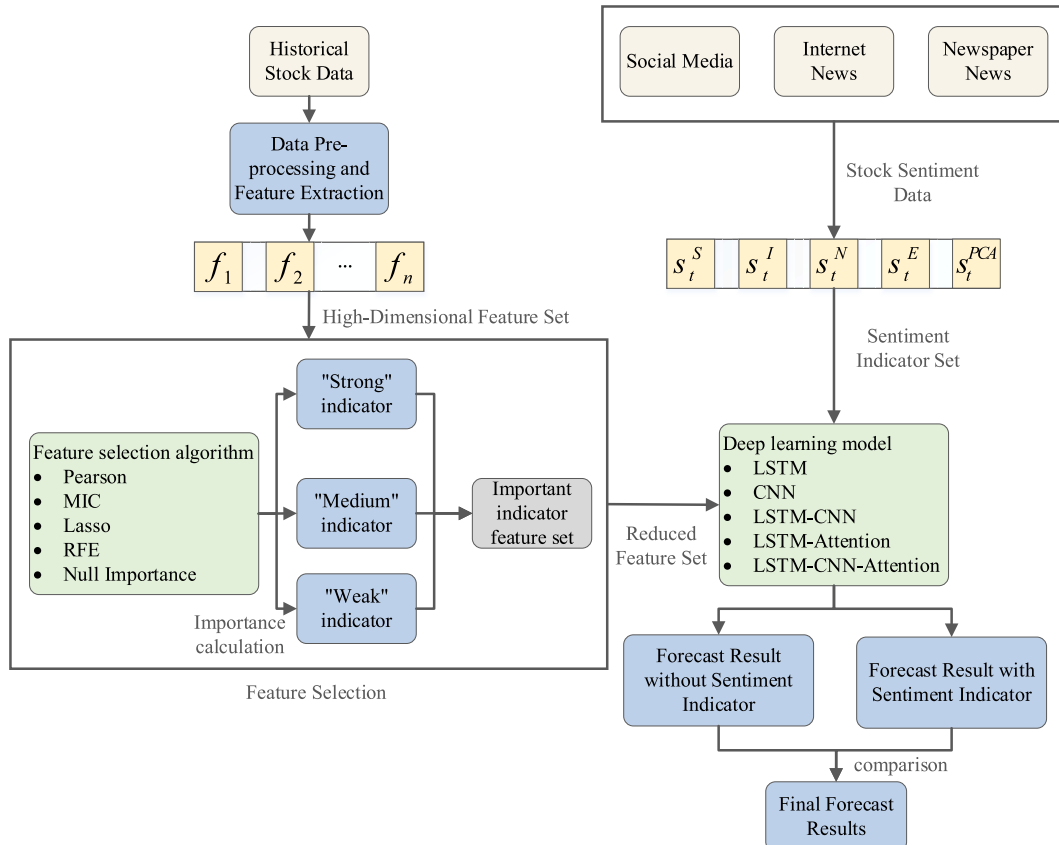


Fig. 1. Overall process of the research.

and Y . Then the MIC values are obtained by Eq 3 and Eq 4.

$$I(X, Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \lg \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

$$m_{x,y} = \frac{\max\{I(x, y)\}}{\lg(\min\{x, y\})}$$

$$MIC = \max(m_{x,y})$$

MIC can detect the nonlinear interaction between features and make up for the limitations of Pearson.

(3) Lasso regression. This algorithm prevents overfitting mainly by constructing a penalty function and regularizing it, and is more suitable for variable selection. The penalty function is represented by the following Eq 5:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (y^{(i)} - (\omega x^{(i)} + b))^2 + \lambda \|\omega\|_1 = \frac{1}{2} MSE(\theta) + \lambda \sum_{i=1}^n |\theta_i|$$

Where m is the number of samples, n is the number of features, ω and θ are vectors of length n and $n+1$ respectively, b is the constant, λ is the constant coefficient, and $\|\omega\|_1$ is the L1 paradigm.

Lasso regression dynamically controls the sparsity of features with λ parameters to eliminate pseudo correlation features.

(4) RFE. The model is constructed through several iterations to screen the features. After screening out the quality features in each iteration, the remaining features continue to the next iteration until all the features are traversed, and the order in which the features are screened out represents the advantages and disadvantages of the features, while whether they are quality features is judged by setting a certain threshold.

The fast primary screening of Pearson and MIC and the iterative optimization of RFE form a workflow, which first eliminates some low-correlation features, and then selects the best 20-30 core features through RFE.

(5) Null Importance. Successfully find and eliminate the features that are highly discriminant or full of noise. Run the model on the original dataset to obtain feature importance, shuffle the labels multiple times, obtain the feature importance under false labels after each shuffle, calculate the difference in feature importance between true and false labels, and filter features based on the difference.

By comparing the Importance of real features and random permutation features, Null Importance effectively removes noise features such as extreme transaction volume and important features that are misjudged by other algorithms.

The five-algorithm selection system in this study achieves the coverage of linear and nonlinear relations, the dual verification mechanism ensures the feature purity, and the combined weight adjustment can adapt to different industry characteristics.

3.3. Sentiment indicator construction

In this study, social media, Internet news and newspaper news were selected as data sources to construct sentiment indicators. Social media reflects the real-time mood swings of retail investors, Internet news reflects the professional interpretation of institutional investors, and newspaper news represents policy-oriented information. Three types of media cover market participants with different granularity, and can capture multiple levels of market sentiment through weighted fusion.

Three sentiment indicators were designed based on the positive and negative semantics of information published on three different media platforms: social media, Internet news and newspaper news, including social media sentiment indicator S_t^S , Internet news sentiment indicator S_t^I and newspaper news sentiment indicator S_t^N . (Liang et al., 2020) The calculation of sentiment indicators is represented by the following formula Eq 6:

$$S_t = \ln \frac{1 + pos_t}{1 + neg_t}$$

Where pos_t is the number of positive comments or news on day t and neg_t is the number of negative comments or news on day t .

In addition, the three sentiment indicators are combined to construct two new sentiment indicators. Firstly, equal combinations are used to design equal sentiment indicator S_t^E based on S_t^S , S_t^I and S_t^N , its calculation is simple and avoids subjective weight bias, but also ignores media influence differences. which is represented by the following Eq 7:

$$S_t^E = \frac{1}{3} S_t^S + \frac{1}{3} S_t^I + \frac{1}{3} S_t^N$$

Next, the first principal component, denoted as S_t^{PCA} , was extracted from the three sentiment indicators S_t^S , S_t^I and S_t^N using the principal component approach and is represented by the following Eq 8:

$$S_t^{PCA} = a_S S_t^S + a_N S_t^N + a_I S_t^I$$

Where a_S , a_N and a_I represent the contribution rate of S_t^S , S_t^N and S_t^I respectively, and the value range is $[-1, 1]$. Principal component approach combines data drive to maximize information retention and meet the dimensional reduction requirements of high-dimensional features.

3.4. Deep learning model

LSTM.

LSTM is a special type of recurrent neural network that can use time series to analyze inputs. Compared with traditional RNN, LSTM introduces a gating mechanism to solve the problem of gradient vanishing or gradient explosion that RNN are prone to encounter when dealing with long-term dependent tasks.

The key to the LSTM is its internal memory unit, consisting of forgetting gates, input gates, and output gates, which remembers and stores key information from the input sequence and decides when and how to pass this information on to the next time step. The forgetting gate is used to decide how to update and retain past memories. Input gates are used to control when and how new input information is fused into the memory unit. Output gates then decide how the memory information extracted at the current time step is passed on to the next time step and the output layer. These gates control the flow of information by learning the weights obtained.

By introducing a gating mechanism, LSTM is able to better capture the long-term dependencies in the input sequence, and has achieved significant performance improvements in tasks such as time series processing and natural language processing. It has been widely used in text generation, machine translation, speech recognition and other fields, and has become one of the important models in deep learning.

CNN.

CNN is a deep learning model for image processing and pattern recognition. The basic idea of CNN is to extract local features in the input data through convolutional operations and gradually combine these features by stacking them in a layer-by-layer fashion to generate more advanced representations.

CNN includes a convolutional layer, a pooling layer and a fully connected layer. In the convolutional layer, one-dimensional convolutional operations are applied to temporal data to extract local temporal features. The pooling layer, on the other hand, is used for down sampling to reduce the computational complexity. Finally, the fully connected layer maps the extracted features to the final output space for classification, regression, or other tasks.

CNN has several advantages in processing time-series data. First, through one-dimensional convolutional operations, it can automatically learn local patterns and long-term dependencies in time-series data, which can better capture features at different time scales. Second, CNN

time-series has the property of parameter sharing, which reduces the number of parameters in the model and improves the efficiency and generalization ability of the model. In addition, by stacking multiple convolutional and pooling layers, CNN tensors can gradually extract more advanced temporal features.

CNN temporal sequences have achieved remarkable results in the fields of audio processing, speech recognition, bioinformatics, natural language processing, etc., and have become one of the most important tools for temporal data analysis.

Attention mechanism.

The Attention Mechanism allows the model to selectively focus on the important elements of different input elements by assigning weights to them. These weights are typically obtained through learning and can be dynamically adjusted according to the model's current state and task requirements. At each time step or level of the model, the Attention Mechanism generates weighted input representations by calculating a weighted sum of the Attention Weights and the inputs, thus directing attention to the important information.

The advantage of the attention mechanism is that it captures the interrelationships and importance of input elements, thus allowing models to process complex tasks and data more accurately. It improves model performance, generalization and interpretability, and can help models to handle long sequences, variable-length inputs and large amounts of data. Its initial application was in natural language processing tasks such as machine translation and language generation, but it is now widely used in computer vision, speech recognition and other areas.

3.5. Evaluation approach

MAE, MSE and RMSE were chosen as evaluation criteria. The calculation of each index is shown in Eqs 9 to 11:

$$MAE = \frac{1}{n} \sum_{i=1}^n |forecast(i) - actual(i)|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (forecast(i) - actual(i))^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (forecast(i) - actual(i))^2}$$

Where $forecast(i)$ is the predicted closing price of a stock on a given day, $actual(i)$ is the actual closing price of a stock on a given day, and n is the number of test datasets. The smaller the value of the above three indicators, the more accurate the model prediction.

4. Experiments and results

4.1. Data collection

The basic stock data of this study comes from China's A-share market, covering 5,164 listed stocks from 2018 to 2022. These data are collected and compiled by Shanghai Rouwei Information Technology Co. These data are not only comprehensive, but also in-depth, including a large number of stock trading characteristics indicators, which cover basic information, market transactions, financial performance, profitability, operational efficiency and other dimensions. Through strict screening of these indicators, key indicators that have a significant impact on stock performance can be identified, thus providing a solid foundation for subsequent research.

The selected time span is five years, during which the market has experienced a variety of changes in the economic environment, including major global events such as the US-China trade war and COVID-19. These events have had a significant impact on market sentiment and stock behavior. As a result, data from this time period can

provide a rich learning material for the model, allowing it to capture the dynamic changes in stock behavior in different market environments. This cross-cycle data collection not only helps to deeply understand the behavior pattern of the stock market, but also provides a valuable information resource for evaluating and predicting the future trend of the stock market.

The investor sentiment data is taken from the Chinese Research Data Services Platform (CNRDS) database, an open and comprehensive data platform for economic, financial and business research in China. In this study, China Listed Companies Financial News Database (CFND) and China Listed Companies Stock Review Database (GUBA) from 2018 to 2022 were used in the CNRDS database. As the first comprehensive big data platform in China that uses advanced artificial intelligence technology to aggregate, summarize and analyze financial news of listed companies, CFND mainly covers two core modules: online financial news and newspaper financial news. In the online financial news module, news reports from more than 400 mainstream online media are gathered. In addition, the platform incorporates more than 400 other important large websites, industry portals and local news sites. The newspaper and financial news module gather the news data of more than 500 well-known newspapers and media, which is widely used as an important data resource for domestic economic, management and business research. At the same time, the module also includes central newspapers, local morning, daily, evening newspapers and various financial and economic newspapers. GUBA focused on China's most influential stock exchange forum, EastMoney.com, where netizens posted a large number of posts about listed companies.

Through text mining and quantitative statistics of the text in the CNRDS database, investors' discussions and views on listed companies and their stocks can be obtained. These discussions and opinions are systematically analyzed and collated into investor sentiment data. These data not only capture the psychological state and expectations of market participants, but also provide a unique perspective for understanding market dynamics. By constructing sentiment indicators, we can more accurately measure and predict the impact of market sentiment on stock price fluctuations, so as to provide strong data support for investment decisions.

4.2. Data pre-processing

There are a total of 425 fields in the 2018-2022 base database, after removing fields that were added midway or deleted midway, fields that are not relevant to the forecast, such as the listing date (listDate), the date of the earnings forecast announcement (forecastNoticeDate), etc., and fields with too many missing values. Since the opening price, the high price, the low price, and the previous day's closing price are themselves closely related to the closing price we are forecasting, this study excludes the above four indexes from the forecasting process, leaving a total of 268 fields. Among them, the trade date index (tradeDate), the stock code index (stockCodeEx), the stock name (securityName), the level 1 of the 2014 edition of the Shenwan industrial classification (induCategoryName shenwan2014_lv1) are the basic information fields, and the closing price (close) is the prediction target.

The 2018-2022 CFND database contains a total of 4,918 stocks in the Internet financial news module, 4,905 stocks in the newspaper financial news module, and 5,024 stocks in the GUBA database. By removing mid-list and mid-list stocks and retaining stocks with complete data in the base, CFND, and GUBA databases for five years, a total of 2,632 stocks remain.

4.3. Selection of stock trading indicators

After preliminary screening, a total of 268 indicators were retained in this study, some of which and their classification are shown in Table 1.

In this study, the importance of each feature is calculated by a

Table 1

Example of indicator classification.

Category of indicators	Field name	Field meaning
Market Transactions	actBuyVol	Outer Trading Volume
	actSellVol	Inside Trading Volume
	freeLiqMv	Free-float Market Capitalization
Financial Performance	annNRoa	Annualized Net Interest Rate on Total Assets
	annRoa	Annualized Rate of Return on Total Assets
	annRoe	Annualized Return on Total Assets
Profitability	dEps	Earnings per share EPS (net/period diluted share capital)
	dRoeAvg	Return on Equity ROE(Net/Average)
	epsTtm	Earnings Per Share (TTM)
Operational Efficiency	faTurnR	Turnover of Fixed Assets
	grsCost2GrRevTtm	Total Operating Cost/Total Operating Revenue (TTM)
	grsCostTtm	Total Operating Cost (TTM)

combined model based on multiple feature selection algorithms, and the calculated importance values are classified using a three-quartile approach. Indicators with importance values between the 2/3 quartile and the maximum value are classified as “Strong (S)” indicators; those with importance values between the 1/3 and 2/3 quartiles are classified as “Medium (M)” indicators; and those with importance values less than the 1/3 quartile are classified as “Weak (W)” indicators. The “weak” indicators are eliminated and the “strong” and “medium” indicators are retained to obtain the important indicator feature set. For the filtered important features, if the feature is “strong” or “medium” four or more times, the feature selection algorithm considers it important and retains it, otherwise it is considered unimportant and will not be considered in the further research.

Among the 2632 stocks, ten stocks were randomly selected for the multiple feature selection algorithm operation, which was repeated three times. According to the experimental results, the importance of 263 indicators was calculated by the combination model of multiple feature selection algorithms, and according to the calculated importance value, it was classified into “strong (S)”, “medium (M)” and “weak (W)”. A total of 26 features were “strong” or “medium” four or more times. Therefore, the feature selection algorithm considers these 26 features as important. The partial results of the feature selection are shown in Table 2.

4.4. Selection of investor sentiment indicators

The principal component analysis method was used to extract the first principal component from social media sentiment indicator S_t^S , Internet news sentiment indicator S_t^I and newspaper news sentiment indicator S_t^N , and the contribution rates of S_t^S , S_t^I and S_t^N were a_S , a_N and a_I . The experimental results showed that the value of a_S was 0.466, the value of a_N was 0.671 and the value of a_I was 0.707. The constructed principal component sentiment indicator S_t^{PCA} was represented by Eq 12:

$$S_t^{PCA} = 0.466S_t^S + 0.671S_t^N + 0.707S_t^I \quad (12)$$

Among them, the contribution rate of Internet news sentiment indicator S_t^I is 70.7%, which is the highest among the three indicators, indicating that it plays a dominant role in the formation of the first principal component of S_t^I . Newspaper news sentiment indicator S_t^N also has a relatively high contribution rate, with a value of 67.1%, and social media sentiment indicator S_t^S contributes 46.6%, which plays a relatively important role. But influence is lower than the other two measures, likely due to the fact that emotional expression on social media differs in intensity and consistency from other media channels. For example, Internet news and newspaper news are more likely to report events with a clear emotional bent, while emotions on social media are more fragmented and diverse.

There are five sentiment indicators included in the prediction comparison experiment in this study, which are: social media sentiment indicator S_t^S , Internet news sentiment indicator S_t^I , newspaper news sentiment indicator S_t^N , equal sentiment indicator S_t^E and principal component sentiment indicator S_t^{PCA} .

4.5. Model evaluation

To prove the validity and applicability of this study, 1030 stocks in 10 industries including pharmaceutical and biological, leisure service, food and beverage, building materials, media, electronics, textile and garment, non-banking finance, computer and transportation were selected for prediction. Among them, 80% of the trading day data is used as a training set to train the model, and the remaining 20% is used to test the model and verify the effectiveness of the model. Five deep learning models, LSTM, CNN, LSTM-CNN, LSTM-Attention and LSTM-CNN-Attention, were used to set comparative experiments using only technical indicators and technical indicators combined with five emotional indicators respectively. The experimental results are shown in Tables 3–12, Where O represents the case where the sentiment indicator is not incorporated. It can be seen from the table that the technical indicators selected after the combination feature selection have the ability to predict the stock price. After adding the sentiment indicator constructed by principal component method, LSTM-CNN-Attention model has better overall prediction performance than other models.

Among the 10 industries, the LSTM-CNN-Attention model performed best. The 7 industries of pharmaceutical and biological, leisure service, food and beverage, media, electronics, non-banking finance and transportation showed the best MAE, MSE and RMSE after integrating the sentiment indicator constructed by the principal component method, while the building materials and computer industries showed the best MAE after incorporating the sentiment indicator constructed by the principal component method, and showed the best MSE and RMSE after integrating the sentiment indicator of social media. The textile and garment industry showed the best MAE after integrating social media sentiment indicator, and showed the best MSE and RMSE after integrating the sentiment indicator constructed by the principal component

Table 2

Importance of partial indicators.

Indicator	Pearson	Imp	MIC	Imp	Lasso	Imp	RFE	Imp	Null Importance	Imp
intLiab2Ev	0.84911	S	0.706762	W	0.026021	S	0.004265	S	0.009852	S
tbAsset	0.74402	S	0.892141	S	2.69E-11	M	0.002656	S	0.008278	S
grsPtMrgnSq	0.736975	S	0.880454	S	0	W	0.015622	S	0.014037	S
opCostTtm	0.708906	S	0.886946	S	6.78E-11	M	0.010584	S	0.003223	S
intLiab	0.706282	S	0.883404	S	9.92E-11	M	0.006479	S	0.006548	S
fcfe	0.704638	S	0.886845	S	1.85E-10	M	0.006876	S	0.014787	S
faTurnR	0.703829	S	0.880513	S	0	W	0.017134	S	0.020082	S
ev2Ebitda	0.680575	S	0.556590	W	0.000151	S	0.012072	S	0.014624	S
opPtTtm	0.672368	S	0.882548	S	1.03E-09	S	0.001145	M	0.011404	S
dNetPrTyoy5	0.657318	S	0.838136	W	0.000379	S	0.005894	S	0.002991	S

Table 3
Pharmaceutical and biological industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0456	0.0320	0.0488	0.0338	0.0380	0.0299
	MSE	0.0100	0.0129	0.0454	0.0193	0.0049	0.0024
	RMSE	0.1001	0.1134	0.2130	0.1387	0.0697	0.0485
CNN	MAE	0.0519	0.0322	0.0518	0.0361	0.0394	0.0369
	MSE	0.0113	0.0065	0.0198	0.0063	0.0047	0.0332
	RMSE	0.1065	0.0808	0.1406	0.0719	0.0689	0.1822
LSTM-CNN	MAE	0.0336	0.0258	0.0317	0.0290	0.0320	0.0255
	MSE	0.0174	0.0274	0.0025	0.0022	0.0025	0.0218
	RMSE	0.1320	0.1657	0.0500	0.0464	0.0503	0.1476
LSTM-Attention	MAE	0.0295	0.0269	0.0289	0.0282	0.0297	0.0223
	MSE	0.0053	0.0020	0.0032	0.0020	0.0067	0.0039
	RMSE	0.0731	0.0450	0.0566	0.0452	0.0817	0.0626
LSTM-CNN-Attention	MAE	0.0230	0.0191	0.0228	0.0222	0.0182	0.0166
	MSE	0.0035	0.0022	0.0036	0.0034	0.0030	0.0021
	RMSE	0.0588	0.0472	0.0599	0.0582	0.0544	0.0459

Table 4
Leisure service industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0356	0.0285	0.0328	0.0305	0.0307	0.0224
	MSE	0.0113	0.0095	0.0121	0.0108	0.0104	0.0021
	RMSE	0.1062	0.0972	0.1102	0.1038	0.102	0.0462
CNN	MAE	0.0417	0.0309	0.0406	0.0313	0.0356	0.0303
	MSE	0.0123	0.0111	0.0115	0.0103	0.0089	0.0102
	RMSE	0.1110	0.1054	0.1073	0.1015	0.0945	0.1012
LSTM-CNN	MAE	0.0235	0.0214	0.0270	0.0222	0.0218	0.0208
	MSE	0.0024	0.0015	0.0035	0.0010	0.0016	0.0019
	RMSE	0.0486	0.0382	0.0590	0.0321	0.0406	0.0431
LSTM-Attention	MAE	0.0254	0.0218	0.0253	0.0223	0.0232	0.0212
	MSE	0.0024	0.0010	0.0103	0.0024	0.0012	0.0011
	RMSE	0.0488	0.0320	0.1014	0.0492	0.0349	0.0325
LSTM-CNN-Attention	MAE	0.0221	0.0187	0.0213	0.0195	0.0204	0.0183
	MSE	0.0015	0.0016	0.0017	0.0013	0.0010	0.0010
	RMSE	0.0381	0.040	0.0416	0.0355	0.0322	0.0320

Table 5
Food and beverage industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0276	0.0156	0.0187	0.0157	0.0170	0.0153
	MSE	0.0016	0.0007	0.0009	0.0008	0.0007	0.0009
	RMSE	0.0405	0.0267	0.0297	0.0275	0.0263	0.0306
CNN	MAE	0.0373	0.0288	0.0319	0.0311	0.0350	0.0198
	MSE	0.0029	0.0021	0.0022	0.0020	0.0028	0.0009
	RMSE	0.0535	0.0462	0.0465	0.0450	0.0533	0.030
LSTM-CNN	MAE	0.0151	0.0131	0.0140	0.0127	0.0136	0.0115
	MSE	0.0008	0.0009	0.0008	0.0008	0.0010	0.0005
	RMSE	0.0287	0.0308	0.0278	0.0280	0.0317	0.0231
LSTM-Attention	MAE	0.0149	0.0123	0.0151	0.0124	0.0145	0.0114
	MSE	0.0005	0.0005	0.0007	0.0005	0.0008	0.0003
	RMSE	0.0230	0.0224	0.0269	0.0220	0.0284	0.0170
LSTM-CNN-Attention	MAE	0.0116	0.0107	0.0117	0.0107	0.0113	0.0097
	MSE	0.0003	0.0005	0.0003	0.0003	0.0004	0.0003
	RMSE	0.0170	0.0230	0.0174	0.0170	0.0188	0.0161

method. Compared with the sentiment indicators of Internet news and sentiment indicators of newspaper, the performance of each model has been improved more after the addition of social media sentiment indicators, and the influence of the same sentiment indicators on the performance of each model is not stable.

4.6. Ablation experiments

In order to systematically evaluate the specific contribution of the feature set after feature selection and the constructed sentiment indicators to the model performance, we designed and conducted a series of ablation experiments to quantify the independent effect of each

component on the model's predictive power by removing the specific feature set and sentiment indicators. The experimental results are shown in [Table 13](#) and [Table 14](#).

The experimental results show that the model without feature selection and sentiment indicators is better than the model without feature selection and sentiment indicators. This result suggests that sentiment indicators have important value in our tasks. Sentiment indicators may capture key emotion or attitude information related to target variables in the data, which provides additional semantic clues for the model in the process of model training, thus helping to improve the model's ability to understand the data and predict the accuracy.

Furthermore, the model with feature selection and sentiment

Table 6
Building materials industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0681	0.0546	0.0651	0.0553	0.0546	0.0533
	MSE	0.0179	0.0077	0.0148	0.0178	0.0166	0.0099
	RMSE	0.1339	0.0878	0.1216	0.1332	0.1290	0.0994
CNN	MAE	0.0742	0.0661	0.0711	0.0704	0.0683	0.0608
	MSE	0.0248	0.0175	0.0212	0.0209	0.0205	0.0124
	RMSE	0.1574	0.1321	0.1455	0.1444	0.1431	0.1114
LSTM-CNN	MAE	0.0535	0.0461	0.0527	0.0514	0.0509	0.0402
	MSE	0.0168	0.0109	0.0124	0.0166	0.0152	0.0065
	RMSE	0.1296	0.1044	0.1112	0.1289	0.1234	0.0808
LSTM-Attention	MAE	0.0504	0.0372	0.0440	0.0393	0.0393	0.0371
	MSE	0.0140	0.0044	0.0073	0.0052	0.0047	0.0036
	RMSE	0.1182	0.0665	0.0852	0.0718	0.0684	0.0597
LSTM-CNN-Attention	MAE	0.0376	0.0335	0.0368	0.0342	0.0343	0.0325
	MSE	0.0064	0.0034	0.0036	0.0038	0.0038	0.0038
	RMSE	0.0803	0.0583	0.0596	0.062	0.0616	0.0617

Table 7
Media industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0243	0.0113	0.0125	0.0115	0.0113	0.0109
	MSE	0.0016	0.0011	0.0021	0.0010	0.0015	0.0016
	RMSE	0.0406	0.0339	0.0462	0.0313	0.0391	0.0394
CNN	MAE	0.0264	0.0234	0.0258	0.025	0.0244	0.0123
	MSE	0.0018	0.0015	0.0017	0.0017	0.0019	0.0009
	RMSE	0.0429	0.0392	0.0416	0.0417	0.0432	0.0305
LSTM-CNN	MAE	0.0110	0.0087	0.0103	0.0099	0.0091	0.0087
	MSE	0.0011	0.0007	0.0012	0.0011	0.0009	0.0011
	RMSE	0.0337	0.0266	0.0351	0.0338	0.0292	0.0332
LSTM-Attention	MAE	0.0104	0.0095	0.0107	0.0104	0.0097	0.0091
	MSE	0.0013	0.0010	0.0012	0.0011	0.0011	0.0009
	RMSE	0.0362	0.0311	0.0346	0.0338	0.0332	0.0298
LSTM-CNN-Attention	MAE	0.0085	0.0079	0.0082	0.0082	0.0081	0.0075
	MSE	0.0007	0.0005	0.0008	0.0007	0.0006	0.0004
	RMSE	0.0257	0.0229	0.0281	0.0267	0.0244	0.0211

Table 8
Electronics finance industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0954	0.0789	0.0894	0.0800	0.0822	0.0647
	MSE	0.0253	0.0207	0.0261	0.0200	0.0213	0.0154
	RMSE	0.1590	0.1438	0.1615	0.1415	0.1460	0.1240
CNN	MAE	0.1445	0.1126	0.1292	0.1272	0.1254	0.0896
	MSE	0.0571	0.0347	0.0396	0.0415	0.0396	0.0252
	RMSE	0.2389	0.1864	0.1990	0.2036	0.1990	0.1586
LSTM-CNN	MAE	0.0724	0.0384	0.0600	0.0524	0.0452	0.0368
	MSE	0.0178	0.0075	0.0129	0.0110	0.0048	0.0077
	RMSE	0.1334	0.0868	0.1138	0.1051	0.0689	0.0880
LSTM-Attention	MAE	0.0442	0.0391	0.0400	0.0397	0.0392	0.0373
	MSE	0.0049	0.0038	0.0047	0.0042	0.0040	0.0111
	RMSE	0.0699	0.0613	0.0682	0.0646	0.0636	0.1052
LSTM-CNN-Attention	MAE	0.0354	0.0301	0.0352	0.0314	0.0297	0.0259
	MSE	0.0035	0.0032	0.0061	0.0063	0.005	0.0029
	RMSE	0.0595	0.0564	0.0778	0.0792	0.0707	0.0538

indicators is better than the model without feature selection but with sentiment indicators. This suggests that feature selection plays an important role in our experiment. The purpose of feature selection is to select the most valuable features for model prediction from a large number of features and remove redundant or irrelevant features. Through feature selection, the model can focus more on important features and reduce the interference of noise, thus improving the generalization ability and prediction performance of the model. When feature selection is not used, the model may be affected by a large number of irrelevant features, resulting in increased model complexity, increased training difficulty, and even overfitting phenomenon, which may affect the prediction effect of the model. After feature selection, the model can

more efficiently use the limited features to learn the rules in the data. Meanwhile, the sentiment indicators, as one of the valuable features, is combined with other selected features to further improve the performance of the model. This shows that reasonable feature selection and the introduction of sentiment indicators are complementary to each other when constructing the model, and the two can enhance the effect of the model to the greatest extent. Fig. 2 shows the MAE values under various conditions when selecting the LSTM-CNN-Attention model. It can be seen that the MAE value is the smallest after feature selection and sentiment indicators is included.

In summary, feature selection can significantly improve the model effect, and the introduction of sentiment indicators further optimizes the

Table 9
Textile and garment industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0316	0.0252	0.0270	0.0256	0.0259	0.0249
	MSE	0.0023	0.0022	0.0019	0.0016	0.0021	0.0019
	RMSE	0.0484	0.0473	0.0433	0.0400	0.0457	0.0441
CNN	MAE	0.0332	0.0280	0.0317	0.0293	0.0309	0.0265
	MSE	0.0025	0.0022	0.0025	0.0023	0.0021	0.0021
	RMSE	0.0503	0.0469	0.0500	0.0475	0.0463	0.0455
LSTM-CNN	MAE	0.0240	0.0200	0.0236	0.0222	0.0224	0.0197
	MSE	0.0020	0.0013	0.0015	0.0014	0.0012	0.0013
	RMSE	0.0447	0.0361	0.0384	0.0369	0.0343	0.0362
LSTM-Attention	MAE	0.0246	0.0172	0.0221	0.0186	0.0189	0.0133
	MSE	0.0021	0.0012	0.0014	0.0016	0.0013	0.0006
	RMSE	0.0454	0.0348	0.0373	0.0400	0.0366	0.0237
LSTM-CNN-Attention	MAE	0.0184	0.0115	0.0120	0.0121	0.0118	0.0121
	MSE	0.0013	0.0006	0.0006	0.0005	0.0005	0.0005
	RMSE	0.0362	0.0238	0.0244	0.0225	0.0216	0.0213

Table 10
Non-banking industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0961	0.0687	0.0956	0.0713	0.0691	0.0665
	MSE	0.0129	0.0122	0.0129	0.0132	0.0121	0.0111
	RMSE	0.1135	0.1104	0.1134	0.1147	0.1099	0.1054
CNN	MAE	0.1039	0.0946	0.0992	0.0972	0.0991	0.0692
	MSE	0.0156	0.0127	0.0134	0.0130	0.0135	0.0123
	RMSE	0.1248	0.1129	0.1157	0.1142	0.1162	0.1109
LSTM-CNN	MAE	0.0659	0.0379	0.0408	0.0411	0.0405	0.0377
	MSE	0.0112	0.0035	0.0044	0.0045	0.0044	0.0038
	RMSE	0.1060	0.0595	0.0664	0.0668	0.0664	0.0620
LSTM-Attention	MAE	0.0449	0.0413	0.0470	0.0428	0.0439	0.0411
	MSE	0.0054	0.0047	0.0057	0.0044	0.0048	0.0044
	RMSE	0.0737	0.0684	0.0757	0.0667	0.0696	0.0663
LSTM-CNN-Attention	MAE	0.0418	0.0347	0.0396	0.0364	0.0377	0.0337
	MSE	0.0046	0.0033	0.0039	0.0036	0.0037	0.0031
	RMSE	0.0681	0.0573	0.0628	0.0598	0.0606	0.0559

Table 11
Computer industry forecast results.

Model		O	$O+S_t^S$	$O+S_t^N$	$O+S_t^I$	$O+S_t^E$	$O+S_t^{PCA}$
LSTM	MAE	0.0439	0.0403	0.0426	0.0409	0.0411	0.0389
	MSE	0.0066	0.0067	0.0057	0.0064	0.0056	0.0039
	RMSE	0.0809	0.0819	0.0754	0.0802	0.0747	0.0627
CNN	MAE	0.0456	0.0414	0.0436	0.0417	0.0426	0.0411
	MSE	0.0047	0.0059	0.0060	0.0043	0.0053	0.0064
	RMSE	0.0688	0.0771	0.0777	0.0658	0.0730	0.0799
LSTM-CNN	MAE	0.0393	0.0295	0.0377	0.0295	0.0323	0.0237
	MSE	0.0055	0.0024	0.0053	0.0024	0.0025	0.0016
	RMSE	0.0745	0.0485	0.0728	0.0486	0.0499	0.0406
LSTM-Attention	MAE	0.0398	0.0310	0.0331	0.0315	0.0345	0.0233
	MSE	0.0050	0.0024	0.0032	0.0024	0.0036	0.0026
	RMSE	0.0704	0.0488	0.0562	0.0493	0.0601	0.0514
LSTM-CNN-Attention	MAE	0.0304	0.0163	0.0222	0.0162	0.0174	0.0153
	MSE	0.0024	0.0012	0.0019	0.0012	0.0013	0.0012
	RMSE	0.0486	0.0342	0.0432	0.0343	0.0363	0.0343

model performance on this basis.

5. Conclusion

This study employs five feature selection algorithms—Pearson correlation coefficient, MIC, Logistic Regression, RFE, and Null Importance—to successfully identify 26 features highly correlated with stock price prediction. These features not only consider traditional market data but also incorporate key factors that influence market volatility. Furthermore, the research calculates three types of sentiment indicators from social media, internet news, and newspaper articles, and combines these using equal combinations and principal component analysis to

construct two composite sentiment indicators. Finally, by integrating these five investor sentiment indicators, a multi-feature fusion model is developed to enhance the accuracy of stock price predictions. To validate the model's effectiveness, the study uses 1030 stocks across 10 industries, including pharmaceutical biotechnology and leisure services, as the research subjects. Experimental results show that the LSTM-CNN-Attention model, integrated with principal component sentiment indicators, demonstrates significant advantages in both feature selection and prediction accuracy, the results of ablation experiments further proved the effectiveness of feature selection and sentiment indicators in improving the performance of the model. This finding not only substantiates the effectiveness of the proposed method in stock market

Table 12
Transportation industry forecast results.

Model		O	O+S _t ^S	O+S _t ^N	O+S _t ^I	O+S _t ^E	O+S _t ^{PCA}
LSTM	MAE	0.0588	0.0516	0.0571	0.0570	0.0544	0.0388
	MSE	0.0103	0.0188	0.0105	0.0103	0.0098	0.0075
	RMSE	0.1016	0.1369	0.1023	0.1014	0.0992	0.0864
CNN	MAE	0.0662	0.0571	0.0601	0.0576	0.0576	0.0549
	MSE	0.0133	0.0093	0.0093	0.0181	0.0216	0.0087
	RMSE	0.1152	0.0967	0.0963	0.1345	0.1469	0.0934
LSTM-CNN	MAE	0.0529	0.0424	0.0520	0.0455	0.0452	0.0409
	MSE	0.0174	0.0073	0.0099	0.0065	0.0087	0.0068
	RMSE	0.1318	0.0855	0.0993	0.0809	0.0933	0.0828
LSTM-Attention	MAE	0.0562	0.0396	0.0496	0.0417	0.0463	0.0394
	MSE	0.0099	0.0079	0.0229	0.0082	0.0089	0.0061
	RMSE	0.0994	0.0888	0.1514	0.0903	0.0946	0.0783
LSTM-CNN-Attention	MAE	0.0459	0.0382	0.0458	0.0379	0.0382	0.0369
	MSE	0.0069	0.0135	0.0068	0.0078	0.0066	0.0066
	RMSE	0.0830	0.1163	0.0823	0.0883	0.0815	0.0812

Table 13
Result of No Feature & No Sentiment.

Model		Ind.1	Ind.2	Ind.3	Ind.4	Ind.5	Ind.6	Ind.7	Ind.8	Ind.9	Ind.10
LSTM	MAE	0.2338	0.2433	0.2268	0.2212	0.2249	0.2561	0.2201	0.2712	0.2713	0.2857
	MSE	0.4154	0.3078	0.1800	0.1664	0.3996	0.4263	0.4215	0.4119	0.3165	0.5306
	RMSE	0.6445	0.5548	0.4243	0.4079	0.6321	0.6529	0.6492	0.6418	0.5626	0.7284
CNN	MAE	0.2733	0.2812	0.2290	0.2468	0.2462	0.2811	0.2245	0.2686	0.2868	0.3010
	MSE	0.5677	0.4112	0.1835	0.2071	0.4789	0.6637	0.4385	0.4040	0.3537	0.5889
	RMSE	0.7534	0.6412	0.4284	0.4551	0.6920	0.8147	0.6622	0.6356	0.5947	0.7674
LSTM-CNN	MAE	0.2197	0.2242	0.2245	0.2013	0.2202	0.2368	0.2003	0.2379	0.2479	0.2625
	MSE	0.4055	0.2614	0.1764	0.1378	0.3831	0.4710	0.3490	0.3169	0.2643	0.2894
	RMSE	0.6368	0.5113	0.4200	0.3712	0.6189	0.6863	0.5908	0.5630	0.5141	0.5380
LSTM-Attention	MAE	0.2176	0.2331	0.1868	0.2458	0.1857	0.2279	0.2032	0.2146	0.2579	0.2411
	MSE	0.4072	0.2825	0.1221	0.5256	0.3311	0.4363	0.3592	0.1918	0.4667	0.2441
	RMSE	0.6381	0.5315	0.3495	0.7250	0.5754	0.6605	0.5994	0.4380	0.6831	0.4941
LSTM-CNN-Attention	MAE	0.1984	0.1917	0.1761	0.2362	0.1646	0.2257	0.1878	0.2112	0.2357	0.2190
	MSE	0.1693	0.1911	0.1085	0.1897	0.2601	0.4279	0.1199	0.1980	0.2860	0.2014
	RMSE	0.4114	0.4371	0.3295	0.4355	0.5100	0.6541	0.3463	0.4450	0.5348	0.4488

* Ind.1: Pharmaceutical and biological; Ind.2: Leisure service; Ind.3: Food and beverage; Ind.4: Building materials; Ind.5: Media; Ind.6: Electronics finance; Ind.7: Textile and garment; Ind.8: Non-banking; Ind.9: Computer; Ind.10: Transportation.

Table 14
Result of No Feature & Sentiment.

Model		Ind.1	Ind.2	Ind.3	Ind.4	Ind.5	Ind.6	Ind.7	Ind.8	Ind.9	Ind.10
LSTM	MAE	0.1093	0.1112	0.0834	0.0923	0.0973	0.1390	0.0678	0.1254	0.1378	0.1434
	MSE	0.0729	0.0643	0.0396	0.0452	0.0615	0.0792	0.0193	0.0881	0.0665	0.0720
	RMSE	0.2700	0.2536	0.1991	0.2125	0.2481	0.2815	0.1389	0.2968	0.2578	0.2683
CNN	MAE	0.1210	0.1134	0.0812	0.1034	0.1015	0.1638	0.0856	0.1478	0.1590	0.1512
	MSE	0.0498	0.0669	0.0376	0.0695	0.0464	0.1100	0.0476	0.1223	0.0885	0.1212
	RMSE	0.2231	0.2586	0.1939	0.2636	0.2153	0.3317	0.2182	0.3498	0.2975	0.3481
LSTM-CNN	MAE	0.0963	0.0897	0.0789	0.0890	0.0878	0.1134	0.0547	0.1034	0.1356	0.1391
	MSE	0.0427	0.0418	0.0355	0.0269	0.0347	0.0836	0.0096	0.0599	0.0791	0.1025
	RMSE	0.2065	0.2045	0.1884	0.1641	0.1863	0.2891	0.0979	0.2447	0.2812	0.3202
LSTM-Attention	MAE	0.0852	0.0764	0.0634	0.0906	0.0734	0.0934	0.0576	0.0878	0.1012	0.1177
	MSE	0.0385	0.0304	0.0141	0.0279	0.0186	0.0567	0.0106	0.0432	0.0440	0.0900
	RMSE	0.1961	0.1742	0.1186	0.1671	0.1364	0.2381	0.1030	0.2078	0.2099	0.3001
LSTM-CNN-Attention	MAE	0.0741	0.0712	0.0527	0.0891	0.0523	0.0912	0.0422	0.0912	0.0834	0.1012
	MSE	0.0236	0.0264	0.0097	0.0270	0.0135	0.0541	0.0057	0.0466	0.0299	0.0666
	RMSE	0.1537	0.1624	0.0986	0.1643	0.1161	0.2325	0.0755	0.2158	0.1729	0.2580

prediction but also highlights the practical significance of sentiment indicators in improving the predictive performance of models.

While this study offers innovative and practical contributions regarding the impact of investor sentiment on stock price prediction, several aspects can be further refined. First, the channels and dimensions for collecting sentiment data can be expanded. Currently, the study relies primarily on sentiment data from social media and news websites, but additional sources, such as international news or forum discussions, could be incorporated to capture a more comprehensive range of global market sentiment fluctuations. Such diversification of data sources

could provide a richer emotional context for the prediction model, enhancing its forecasting capability. Second, the generalization ability of the model can be further improved. The current model is trained primarily using data from a single market, and future research could consider incorporating broader cross-industry and cross-market trading data. By integrating large language models (e.g., GPT-based models) to expand the data sources and conducting multi-dimensional testing and optimization, the model's adaptability in different market environments could be enhanced, particularly in responding to sudden events and providing real-time predictions. Lastly, while the proposed multi-feature

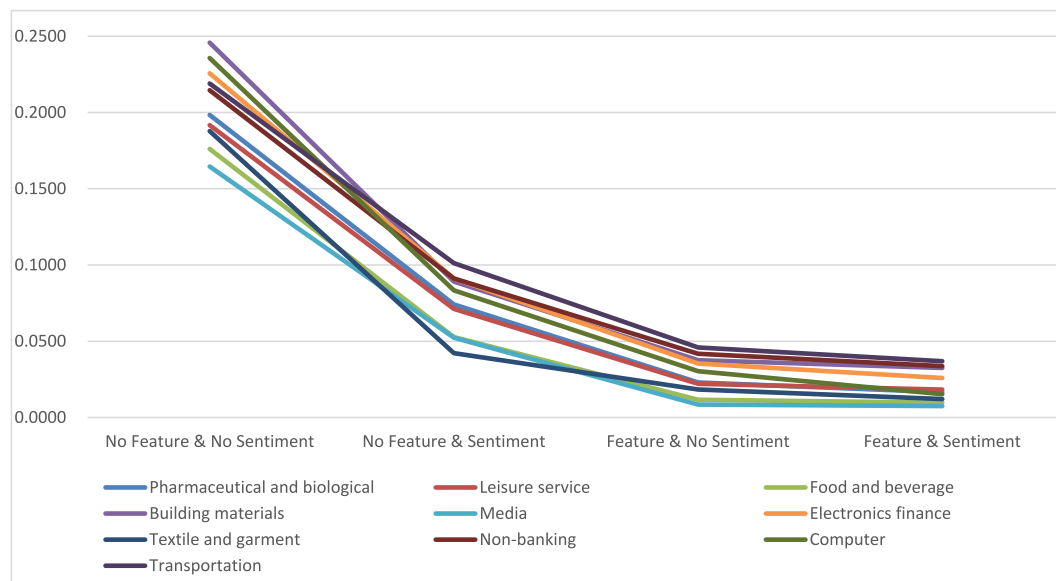


Fig. 2. MAE values for 10 industries (based on LSTM-CNN-Attention).

fusion model has demonstrated notable improvements in prediction accuracy, its interpretability remains an area for further development. Future research will focus on exploring how to clarify the causal relationship between investor sentiment and stock price or index movements, analyzing how sentiment fluctuations trigger price volatility in the stock market, and quantifying the specific impact of sentiment on stock price movements. This would not only enhance the transparency of the model but also provide investors with more targeted decision-making support.

Implementing these improvements is expected to offer investors more precise market insights, helping them make more rational and informed investment decisions in complex financial environments, thereby achieving higher returns and better risk control.

Author contributions

K.H.Z. and D.X. conceived the experiments, K.H.Z. conducted the experiments, wrote the main manuscript text and analyzed the results, D.X. supervised and revised the whole manuscript text, X.C.H provided funding support. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work described in this paper are supported by Guangxi Key Laboratory of Big Data in Finance and Economics (Grant No. FEDOP2022).

Data availability

The data supporting the findings of this study are available in the supplementary Information files and upon reasonable request from the corresponding authors. Code for this study: <https://github.com/zZ1117/Model-code>.

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